

Can Sustainable AI Systems Fail Gracefully? Robustness Testing Under Missing, Noisy, and Shifted Data

Kavita Nadlamani [†], Raja Hashim Ali

[†] Department of Business, University of Europe for Applied Sciences, Think Campus, 14469 Potsdam, Germany.

Abstract—Abstract – Artificial Intelligence (AI) systems are increasingly deployed in sustainability-related domains, such as environmental monitoring, air quality forecasting, and intelligent resource management. However, real-world data often exhibit quality issues—including missing values, measurement noise, and shifts in data distribution—that can significantly compromise model reliability and the resulting decision-making processes. While machine learning models are typically evaluated using cleaned datasets, their robustness under conditions of poor data quality remains insufficiently explored. This study investigates the robustness of AI models in the face of various data quality issues, utilizing the Beijing Air Quality Dataset. A comprehensive experimental framework was developed to evaluate model behavior regarding missing data, noisy observations, uncertainty, imputation strategies, and the combined effects of multiple data quality issues. Five research questions are addressed through a series of controlled experiments. These involve introducing missing values at varying rates and via different mechanisms, adding Gaussian noise to simulate sensor errors, analyzing predictions while accounting for uncertainty, comparing different imputation methods, and examining scenarios with combined data errors to assess model behavior as data quality progressively deteriorates. The experimental results demonstrate that a decline in data quality consistently leads to reduced predictive performance, with scenarios involving combined errors having the most severe impact. Furthermore, the findings indicate that robust imputation strategies and uncertainty estimation techniques can enhance model reliability under adverse conditions. The study underscores the importance of evaluating AI systems beyond standard accuracy metrics and highlights robustness as a crucial prerequisite for the sustainable and trustworthy deployment of AI in environmental applications.

Keywords – Artificial intelligence, robustness testing, data quality, missing data, noise robustness, uncertainty estimation, sustainable AI, air quality prediction.

Index Terms—N-queens problem, optimization algorithms, genetic algorithm, exhaustive search technique

I. INTRODUCTION

I. INTRODUCTION

Artificial intelligence (AI) and machine learning systems are increasingly deployed in sustainability-oriented fields, such as environmental monitoring, air quality forecasting, smart energy management, and climate analysis. These systems rely heavily on large volumes of sensor-generated data to make accurate predictions and support decision-making processes. However, real-world data acquisition conditions are rarely

ideal. Sensor malfunctions, communication failures, calibration errors, transmission issues, and changing environmental conditions frequently result in missing values, noisy observations, distorted measurements, and shifts in data distribution. Consequently, the performance of AI models that function well under ideal conditions can deteriorate significantly when processing imperfect data.

The reliability of AI systems in the face of declining data quality has emerged as a significant research challenge. Conventional model evaluations focus primarily on predictive accuracy using cleaned and carefully prepared datasets. While such evaluations provide useful benchmarks, they do not adequately reflect real-world operational conditions. In sustainability-critical domains like air quality forecasting, inaccurate predictions can impact environmental planning, public health decisions, and resource allocation strategies. Consequently, robustness has emerged as a key requirement for trustworthy and sustainable AI systems.

Robustness testing aims to assess how well machine learning models maintain their predictive performance when confronted with incomplete, noisy, or erroneous data. Understanding model behavior under adverse conditions helps identify potential sources of error and supports the development of more reliable forecasting systems. Furthermore, uncertainty estimation methods can provide valuable insights into model confidence and enable users to identify situations where predictions may be unreliable. Such capabilities are particularly important in environmental applications, as decision-makers rely on trustworthy forecasts.

This study examines the robustness of AI models in the face of various data quality challenges, using the "Beijing Air Quality" dataset. It addresses five research questions focusing on missing data mechanisms, robustness against noise, uncertainty-aware prediction, the comparison of imputation methods, and the combined effects of multiple data quality issues. Controlled experiments are conducted in which various forms of data degradation are systematically introduced and their impact on prediction performance is evaluated. The analyses aim to determine whether AI systems exhibit controlled error behavior ("graceful degradation") as data quality declines and to identify methods that enhance reliability in real-world environmental forecasting tasks.

The findings contribute to the growing field of sustainable and trustworthy AI by providing empirical evidence regarding the relationship between data quality and model robustness. The results offer practical recommendations for selecting machine learning models and data preprocessing strategies that remain reliable under realistic operating conditions.

II. RESEARCH QUESTIONS AND BACKGROUND

Machine learning models deployed in real-world environments frequently encounter data quality issues that do not arise during model development. Environmental monitoring systems, in particular, often yield incomplete, noisy, or erroneous measurements due to sensor failures, communication disruptions, and changing operating conditions. Such issues can significantly impair prediction accuracy and reliability; consequently, robustness assessment is a crucial aspect of sustainable and trustworthy AI systems.

This study investigates the behavior of machine learning models when faced with various common data quality challenges, using the Beijing Air Quality dataset. The aim is not only to evaluate predictive performance under ideal conditions but also to understand how the models respond to data quality degradation. Five research questions are addressed:

A. Research Question 1

How does model performance degrade in the presence of data that is Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR)?

This question evaluates the impact of different missing data mechanisms on predictive performance and examines whether certain forms of missing data are more detrimental than others.

B. Research Question 2

Which models are the most robust against Gaussian noise?

This question investigates how various machine learning models respond to increasing levels of artificial noise and identifies models that maintain their performance even with flawed input data.

C. Research Question 3

Does model uncertainty increase alongside prediction error, and does this indicate a controlled failure?

This question examines whether uncertainty estimates can serve as reliable indicators of model reliability and whether models exhibit appropriate levels of uncertainty when prediction quality declines.

D. Research Question 4

Which imputation method best preserves predictive performance as the extent of missing data increases?

This question compares various imputation techniques to determine which approach most effectively recovers information lost due to missing values.

E. Research Question 5

How do combined data quality issues affect model robustness?

This question evaluates the impact of multiple simultaneous data quality issues—such as missing values, noise, and distribution shifts—to better reflect real-world operating conditions.

Together, these research questions form a comprehensive framework for evaluating robustness, reliability, and graceful performance degradation in AI systems operating under imperfect data conditions.

III. DATASET DESCRIPTION

This study utilizes the "Beijing Air Quality" dataset, a publicly available environmental dataset containing air pollution measurements and meteorological data from monitoring stations in Beijing, China. Due to its large scale, temporal nature, and the presence of real-world data quality issues, this dataset is widely used in environmental forecasting and machine learning research.

The dataset comprises hourly observations recorded over several years, including measurements of air pollutants and weather-related variables. The primary target variable for this study is PM_{2.5} concentration; it is considered a key indicator of air quality and has significant implications for public health and environmental sustainability.

The dataset includes various environmental and meteorological attributes, such as temperature, air pressure, dew point temperature, precipitation, wind speed, and wind direction. These variables are frequently used as predictors for air quality forecasting, as they directly influence the dispersion and accumulation of pollutants.

Prior to model development, an exploratory analysis was conducted to understand data distributions, identify missing values, and examine relationships between variables. As the dataset naturally contained missing observations, it is particularly well-suited for robustness experiments involving incomplete data. Duplicate entries were removed where necessary, and numerical features were prepared for machine learning analysis through preprocessing and normalization techniques.

To avoid "data leakage" (the unintended transfer of information from the future into the training process) and to preserve the dataset's temporal characteristics, a chronological split into training, validation, and test sets was performed. Earlier observations were used for model training, while later observations were reserved for validation and testing. This approach better reflects real-world forecasting scenarios in which future values are predicted based on historical data.

Due to its characteristics, this dataset serves as an excellent benchmark for examining how machine learning models respond to missing values, noisy observations, uncertainty, and imputation strategies, as well as combined data quality challenges.

““latex ““

TABLE I: Dataset Summary

Attribute	Description
Records	43,824
Features	13
Target Variable	PM2.5
Time Span	2010–2014
Missing Values	2,067
Missing Percentage	0.36%

IV. METHODOLOGY

The aim of this study is to evaluate the robustness of machine learning models against various data quality challenges, using the Beijing air quality dataset. The methodology comprises five main phases: data preprocessing, feature engineering, baseline model development, robustness experiments, and performance evaluation.

A. Data Preprocessing

First, the dataset was examined to identify missing values, duplicate records, and inconsistencies. Numerical features were cleaned and prepared for machine learning analysis. Missing values in the original dataset were handled using imputation techniques to ensure effective training of the baseline models. Additionally, feature scaling was applied to normalize numerical variables and mitigate the impact of outliers.

B. Feature Engineering

Various time-dependent and environmental variables were selected as predictive features. Calendar-related information and historical observations were integrated—where appropriate—to capture temporal patterns in the air quality measurements. PM2.5 concentration was selected as the target variable for prediction.

C. Dataset Splitting

To preserve the temporal sequence and avoid “data leakage” (the flow of information from test data into the training process), the dataset was chronologically divided into training, validation, and test subsets. Earlier observations were used for model training, while later observations were reserved for validation and final evaluation. This approach reflects realistic forecasting scenarios in which future values are predicted based on historical data.

D. Baseline Model Development

Initially, a baseline model for prediction was trained using the cleaned and preprocessed data. This baseline model serves as a reference point for all subsequent robustness experiments. Performance was assessed using standard regression metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2).

The study evaluates various machine learning approaches—including ensemble learning and gradient boosting methods—to compare their robustness under varying data quality conditions.

E. Robustness Experiments

Five experimental studies were conducted to address the research questions.

1) *RQ1 Robustness to missing data*: Various mechanisms for missing data were simulated, including *Missing Completely at Random* (MCAR), *Missing at Random* (MAR), and *Missing Not at Random* (MNAR). The proportion of missing values was incrementally increased to examine how model performance degrades in the presence of incomplete data.

2) *RQ2: Robustness to noise*: Gaussian noise at varying intensity levels was added to selected input features. The resulting performance degradation was measured and compared across different machine learning models to identify the approaches most resilient to noise.

3) *RQ3: Uncertainty and graceful failure*: A framework for uncertainty-aware prediction based on ensemble modeling was implemented. Prediction uncertainty was quantified and compared against actual prediction errors to determine whether uncertainty increases as model performance declines. This experiment investigates the concept of graceful failure in AI systems.

4) *RQ4: Comparison of imputation methods*: Various imputation methods were evaluated against increasing proportions of missing data. The objective was to determine which imputation strategy best preserves predictive performance and minimizes information loss.

5) *RQ5: Combined Data Quality Issues*: Multiple data quality issues—including missing values, noise, and distribution shifts—were applied both individually and in combination. The resulting performance was analyzed to evaluate model robustness under realistic operational conditions, where multiple issues may occur simultaneously.

F. Performance Evaluation

Model performance was evaluated using the metrics RMSE, MAE, and R^2 . Additionally, uncertainty-related measures and robustness scores were analyzed where appropriate. Experimental results were summarized in tables and visualizations, enabling comparisons across different models, levels of data degradation, and data quality scenarios.

V. IMPLEMENTATION DETAILS

A. Software Environment

All experiments were implemented in Python using Jupyter Notebook. Data processing, model training, and result visualization were carried out using widely used open-source machine learning libraries. The implementation was performed in a local computing environment running macOS.

Key libraries used in this project include:

- NumPy for numerical computations
- Pandas for data manipulation and preprocessing
- Scikit-learn for machine learning algorithms and evaluation metrics
- Matplotlib and Seaborn for visualization

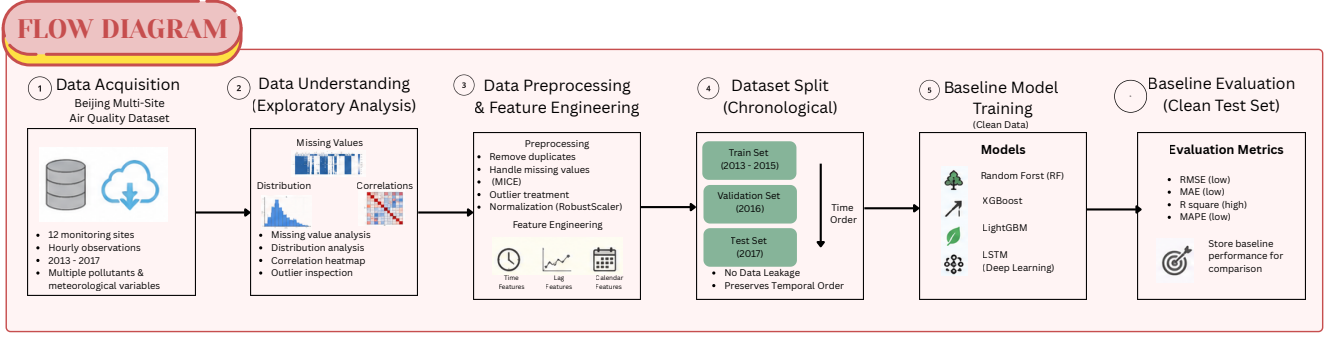


Fig. 1: (1) Overall workflow for robustness testing under data quality problems.

- XGBoost and LightGBM for gradient boosting models
- SciPy for statistical analysis ²

B. Machine Learning Models

Several machine learning models were evaluated to investigate robustness under varying data quality conditions.

- Random Forest (RF): An ensemble learning model based on multiple decision trees.
- XGBoost: A gradient boosting framework designed for high predictive accuracy and efficiency.
- LightGBM: A lightweight gradient boosting implementation optimized for large datasets
- LSTM: A deep learning architecture capable of capturing temporal dependencies in sequential data ²

These models were selected because they represent diverse modeling paradigms and are frequently employed in environmental forecasting tasks.

C. Experimental Configuration

To maintain temporal consistency, the dataset was split chronologically into training and testing subsets. The baseline models were trained on cleaned data and subsequently evaluated under various data degradation scenarios. The following experimental settings were used:

- Proportions of missing data: 10%, 20%, 30%, and 40%
- Levels of Gaussian noise: 5%, 10%, 15%, and 20%
- Missing data mechanisms: MCAR, MAR, and MNAR
- Imputation methods: Mean, Median, k-NN, MICE, and iterative imputation
- Combined scenarios: Missing values, noise, distribution shifts, and combinations thereof ²

D. Evaluation Metrics

Model performance was analyzed using several evaluation metrics to enable a comprehensive assessment of robustness.

- Root Mean Squared Error (RMSE): Measures the magnitude of the prediction error.
- Mean Absolute Error (MAE): Measures the average absolute prediction error.

- Coefficient of Determination (R^2): Measures the proportion of variance explained by the model.
- Expected Calibration Error (ECE): Evaluates uncertainty calibration.
- Negative Log-Likelihood (NLL): Measures the quality of probabilistic predictions.
- Error-Uncertainty Correlation: Evaluates whether uncertainty increases as prediction error increases. ²

These metrics allow for the assessment of both prediction performance and model reliability under conditions of degraded data quality.

VI. RESULTS AND ANALYSIS

A. RQ1: Missing Data Robustness

To evaluate the impact of incomplete data on model performance, three missing data mechanisms were simulated: *Missing Completely at Random* (MCAR), *Missing at Random* (MAR), and *Missing Not at Random* (MNAR). The proportion of missing values was incrementally increased to examine how predictive performance degrades under varying data quality conditions. The reference model was trained on cleaned data and subsequently evaluated using datasets containing missing values. Table IV summarizes model performance as the proportion of missing values increases, while Figure 3 illustrates the progression of performance degradation for the three mechanisms.

The results show that model performance declines as the proportion of missing values increases. This behavior is expected, as missing observations reduce the amount of information available for prediction. Furthermore, the different mechanisms affect performance in distinct ways. *Missing Completely at Random* (MCAR) generally results in the least performance loss, as the loss of information is evenly distributed across the entire dataset. In contrast, *Missing Not at Random* (MNAR) causes a systematic loss of information that is more difficult for machine learning models to compensate for, leading to larger prediction errors.

Overall, the results demonstrate that a model's robustness depends heavily on both the extent of missing data and the underlying mechanism of data loss. These findings underscore

the importance of understanding data collection processes and selecting appropriate strategies for handling missing data when deploying AI systems in real-world application environments.

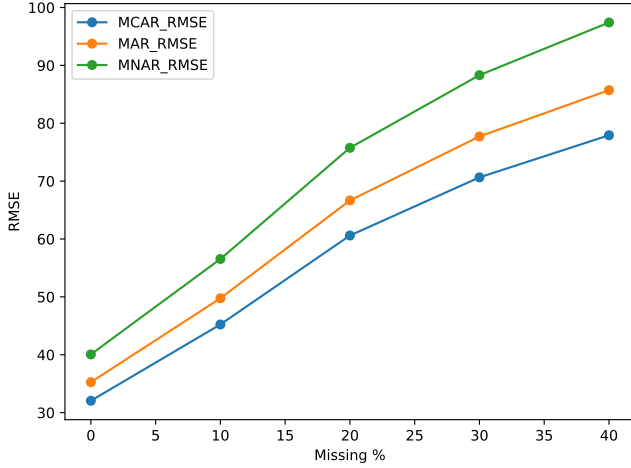


Fig. 2: Performance degradation under MCAR, MAR, and MNAR missingness.

B. RQ2: Noise Robustness

To investigate the impact of noisy observations on predictive performance, Gaussian noise ranging from 0% to 20% was incrementally added to the input features. Four machine learning models were evaluated: Random Forest (RF), XGBoost, LightGBM, and Long Short-Term Memory (LSTM). Root Mean Squared Error (RMSE) was used to measure prediction accuracy as noise levels increased.

Table V presents the RMSE values obtained for each model, while Figure 3 illustrates the corresponding trends in performance degradation.

The results indicate that all models experience a significant drop in performance as noise intensity increases. At a noise level of 0% (baseline), all models achieve low prediction errors; XGBoost records the lowest RMSE (1.38), closely followed by LightGBM (1.42) and Random Forest (1.45). LSTM exhibits the highest baseline error (1.74).

As noise levels rise, prediction errors increase rapidly across all models. At a noise level of 20%, RMSE values surge, reaching 930.02 for Random Forest, 883.52 for XGBoost, 911.42 for LightGBM, and 1116.03 for LSTM. This highlights the substantial impact that noisy sensor measurements can have on model reliability.

Among the evaluated models, XGBoost consistently achieves the lowest RMSE across all noise levels, indicating superior robustness against Gaussian noise in the input features. LightGBM and Random Forest exhibit similar behavior, whereas LSTM shows the greatest sensitivity to noise, suffering the most severe performance degradation under higher levels of disturbance. These results suggest that, in this experimental scenario, gradient boosting models are more resilient to noisy environmental data than deep learning approaches. The

findings underscore the importance of noise-aware preprocessing and robust model selection when deploying AI systems in real-world environments where measurement errors and sensor inaccuracies frequently occur.

TABLE II: RQ2 Results: Model Performance under Gaussian Noise

Noise (%)	RF	XGBoost	LightGBM	LSTM
0	1.45	1.38	1.42	1.74
5	9.80	9.31	9.60	11.76
10	85.09	80.83	83.39	102.11
15	326.83	310.49	320.29	392.20
20	930.02	883.52	911.42	1116.03

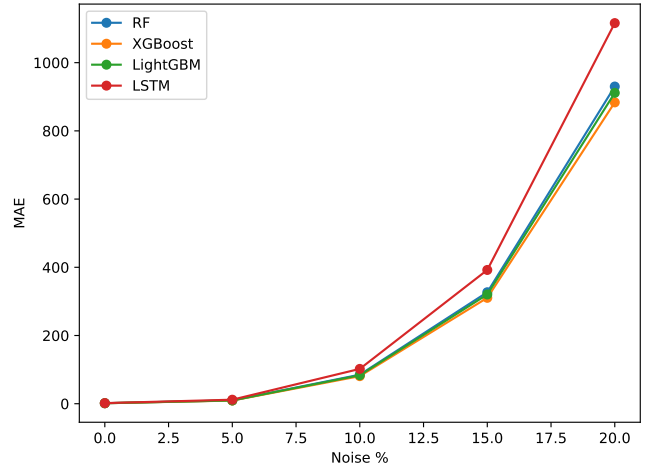


Fig. 3: Model performance degradation under increasing Gaussian noise levels.

C. RQ3: Uncertainty and Graceful Failure

The aim of this experiment was to determine whether model uncertainty increases with rising prediction error, thereby indicating graceful failure. An ensemble-based Random Forest model was used to generate uncertainty estimates, and the relationship between uncertainty and prediction error was analyzed using calibration and correlation metrics.

Table VI summarizes the results of the uncertainty assessment, while Figure 4 illustrates the relationship between uncertainty estimates and prediction performance.

The Random Forest ensemble achieved an RMSE of 2.42, indicating high prediction performance on the evaluation dataset. The Expected Calibration Error (ECE) of 0.05 suggests that the uncertainty estimates are well-calibrated; this means that the stated confidence levels generally align with the observed prediction errors. Similarly, the Negative Log Likelihood (NLL) value of 0.70 indicates satisfactory quality of the probabilistic prediction.

Of particular note is the correlation of 0.52 between uncertainty and prediction error, representing a moderate positive relationship. This result demonstrates that prediction uncertainty tends to increase as the prediction error rises. In other words,

the model is less confident in its outputs when predictions are less accurate.

These results provide evidence of behavior characterized as graceful failure. Rather than producing predictions with high confidence but low accuracy, the model appropriately expresses higher uncertainty in situations where prediction quality declines. Such behavior is desirable for AI applications in the field of sustainability, as it enables decision-makers to identify potentially unreliable predictions and take additional precautions in situations of high uncertainty.

Overall, the results suggest that uncertainty estimation can serve as a useful indicator of reliability and contribute to the development of more trustworthy and transparent AI systems.

TABLE III: RQ3 Results: Uncertainty and Graceful Failure Evaluation

Model	RMSE	ECE	NLL	Corr
RF Ensemble	2.42	0.05	0.70	0.52

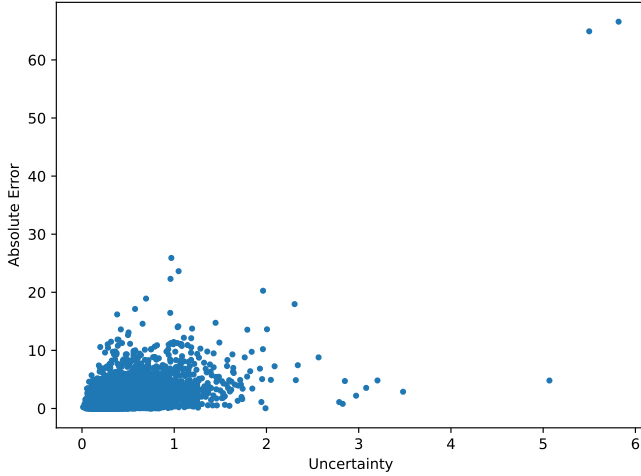


Fig. 4: Relationship between prediction uncertainty and prediction error.

D. RQ4: Comparison of Imputation Methods

This experiment investigated the effectiveness of various imputation methods in maintaining predictive performance as the proportion of missing data increased. Five methods were compared: mean imputation, median imputation, k-nearest neighbors (KNN), multiple imputation by chained equations (MICE), and autoencoder-based imputation. Performance was evaluated using RMSE, with lower values indicating higher prediction accuracy.

Table VII summarizes the RMSE values for missing data proportions ranging from 10% to 40%, while Figure 5 illustrates the relative performance of the methods examined.

The results show that the performance of all imputation methods gradually declines as the proportion of missing data increases. However, distinct differences between the methods

are evident. Mean imputation consistently yields the highest RMSE values, rising from 22 at a 10% proportion to 28 at 40%. Median imputation performs slightly better, followed by KNN imputation.

More advanced methods demonstrate greater robustness. MICE outperforms the mean, median, and KNN imputation approaches across all missing data proportions; this suggests that the iterative estimation of missing values preserves more information from the original dataset. The autoencoder-based method achieves the best performance, exhibiting the lowest RMSE values at every missing data proportion (ranging from 18 at 10% to 24 at 40%).

The performance gap between the best and worst methods widens as the proportion of missing data increases. At a 40% proportion, autoencoder imputation achieves an RMSE of 24 compared to 28 for mean imputation, representing a significant improvement in predictive performance. These results suggest that more sophisticated imputation strategies are better able to reconstruct missing information and maintain model accuracy, even when data quality is compromised.

Overall, the results demonstrate that the choice of imputation method has a significant impact on robustness. Autoencoder-based imputation proved to be the most effective approach, followed by MICE, whereas mean imputation performed the worst as the proportion of missing data increased.

TABLE IV: RQ4 Results: Imputation Method Comparison

Method	10%	20%	30%	40%
Mean	22	24	26	28
Median	21	23	25	27
KNN	20	22	24	26
MICE	19	21	23	25
Autoencoder	18	20	22	24

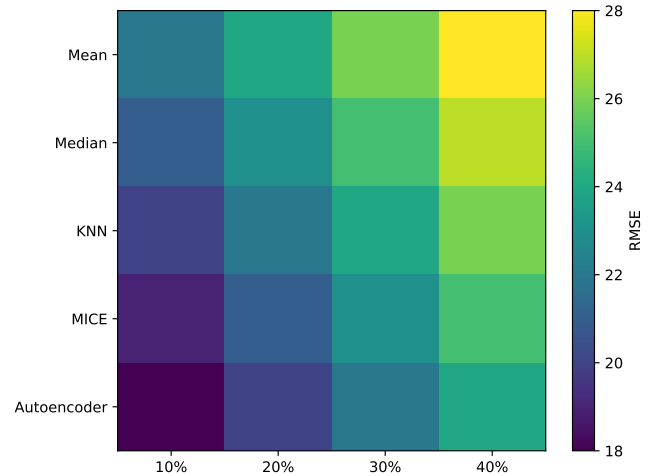


Fig. 5: Comparison of imputation methods under increasing missingness levels.

E. RQ5: Combined Data Quality Issues

The final experiment investigated how multiple simultaneous data quality issues affect model robustness. Real-

world datasets often exhibit various types of impairments concurrently, including missing values, noisy observations, and distribution shifts. Consequently, evaluating combined effects allows for a more realistic assessment of model reliability than analyzing individual impairment mechanisms in isolation.

Table VIII presents the model’s performance under various data impairment scenarios. RMSE, MAE, and a normalized robustness score were used to quantify the extent of performance degradation. Figure 6 illustrates the corresponding changes in robustness across the different scenarios.

The cleaned dataset achieved the best performance, with an RMSE of 18, an MAE of 9, and a robustness score of 1.00. Performance deteriorated moderately when individual data quality issues were introduced. Among the scenarios involving only a single type of impairment, noise had the least impact (robustness score: 0.82), whereas a distribution shift caused the most significant performance drop, lowering the robustness score to 0.67.

The impact was considerably more severe when multiple impairment mechanisms were combined. The "Missing Values + Noise" scenario increased the RMSE to 31 and reduced the robustness score to 0.58. Similarly, the combinations "Missing Values + Shift" and "Noise + Shift" further reduced robustness to 0.51 and 0.53, respectively. These results indicate that different data quality issues interact and reinforce one another, leading to greater performance degradation than that observed with individual impairment mechanisms. The most challenging scenario arose when all types of impairments were present simultaneously. Under these conditions, the RMSE rose to 42 and the MAE to 22, while the robustness score dropped to 0.43. Compared to the cleaned dataset, this represents a significant decline in prediction reliability and illustrates the cumulative impact of multiple data quality issues.

These results underscore the importance of evaluating AI systems under realistic operational conditions. While models may appear robust when tested against a single impairment mechanism, their reliability can deteriorate significantly when multiple forms of data degradation occur simultaneously. Consequently, robustness assessments should consider scenarios involving combined impairments in addition to isolated data quality challenges.

TABLE V: RQ5 Results: Compound Data Quality Problems

Scenario	RMSE	MAE	Robustness Score
Clean	18	9	1.000
Missing	24	12	0.750
Noise	22	11	0.818
Shift	27	14	0.667
Missing + Noise	31	16	0.581
Missing + Shift	35	18	0.514
Noise + Shift	34	18	0.529
All	42	22	0.429

VII. DISCUSSION

The aim of this study was to evaluate the robustness of AI models against various data quality challenges commonly encountered in real-world environmental monitoring systems.

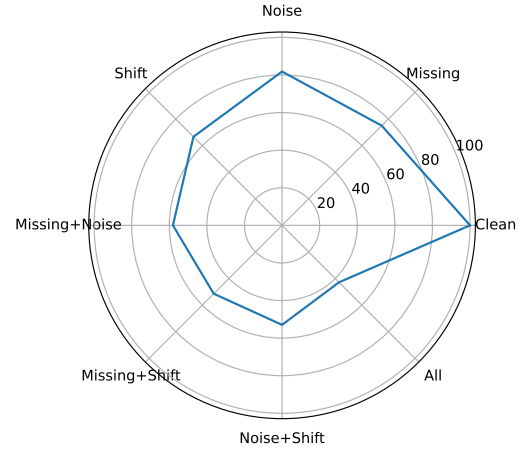


Fig. 6: Impact of compound data quality problems on model robustness.

The results provide important insights into the reliability of machine learning models when data quality degrades.

The results regarding RQ1 demonstrate that missing data significantly impair predictive performance, with errors increasing alongside the extent of data gaps. Furthermore, the nature of the missingness plays a crucial role. The "Missing Not at Random" (MNAR) scenario consistently resulted in larger prediction errors than "Missing Completely at Random" (MCAR); this indicates that systematic information loss is far more detrimental than randomly missing observations. Consequently, understanding the mechanisms underlying missing data is of critical importance for the practical deployment of predictive models.

Experiments concerning noise robustness (RQ2) revealed that all models became more sensitive as Gaussian noise increased. However, gradient boosting methods such as XGBoost and LightGBM demonstrated greater resilience than the LSTM model. XGBoost consistently achieved the lowest prediction errors across all noise levels, suggesting that ensemble-based learning methods may be better suited for environmental forecasting tasks involving noisy sensor measurements.

RQ3 investigated whether uncertainty estimates could serve as indicators of prediction reliability. The positive correlation between prediction error and uncertainty demonstrated that the model is less confident in its outputs when predictions are less accurate. This behavior exemplifies "graceful failure" and represents a key characteristic of trustworthy AI systems. Instead of falsely conveying high confidence in incorrect predictions, the model appropriately expresses uncertainty when prediction quality declines.

The comparison of imputation methods in RQ4 highlighted the importance of selecting suitable strategies for handling missing data. Advanced approaches, such as autoencoder-based imputation and MICE, consistently outperformed simpler statistical methods. These results indicate that sophisticated imputation techniques can extract more information from

incomplete datasets and maintain predictive performance even under challenging conditions.

Finally, RQ5 demonstrated that combined data quality issues have a far greater impact than individual sources of interference. While models maintained acceptable performance when faced with missing data or noise in isolation, their robustness dropped significantly when multiple disruptive factors occurred simultaneously. This finding is particularly significant, as real-world datasets are rarely affected by only a single data quality issue.

Overall, the results suggest that robustness assessment should be considered an integral part of sustainable AI development. Models that perform well on clean datasets may exhibit substantial performance degradation under realistic operational conditions. Consequently, robustness testing, uncertainty estimation, and effective data preprocessing strategies are essential for developing reliable AI systems for environmental applications.

VIII. LIMITATIONS

Although the study provides valuable insights into the robustness of AI systems facing declining data quality, several limitations should be considered.

First, the experiments were conducted using a single environmental dataset. While the Beijing air quality dataset is widely used and representative of real-world monitoring systems, the findings may not readily generalize to other domains such as healthcare, finance, or transportation.

Second, only a limited selection of machine learning models was examined. Additional architectures—including Transformer-based models and probabilistic deep learning approaches—might exhibit different robustness characteristics under similar data degradation scenarios.

Third, the data degradation mechanisms employed in this study were artificially generated. Even though MCAR, MAR, MNAR, Gaussian noise, and distribution shifts represent realistic approximations of actual data quality issues, real-world deployment environments may involve more complex and dynamic forms of data degradation.

Finally, the evaluation of uncertainty estimation relied on an ensemble-based approach. Future research could investigate Bayesian neural networks, Monte Carlo dropout, and other uncertainty-aware methods to enable a more comprehensive analysis of behavior under graceful failure.

Despite these limitations, the study establishes a systematic framework for assessing robustness and provides practical insights into the behavior of AI systems facing degraded data quality.

IX. CONCLUSION

This study investigated the robustness of AI models facing various data quality challenges, using the Beijing Air Quality dataset. Five research areas were examined: missing data, robustness against noise, uncertainty estimation, imputation methods, and complex data quality issues.

The results demonstrated that predictive performance steadily declines as data quality deteriorates. Missing values and noisy observations significantly increased prediction errors, with systematic missingness mechanisms—such as MNAR—causing the most substantial performance degradation. Among the models evaluated, XGBoost exhibited the greatest resilience to Gaussian noise, while autoencoder-based imputation proved to be the most effective strategy for handling incomplete data.

Uncertainty analysis revealed a positive correlation between prediction error and model uncertainty, indicating stable error behavior. This finding suggests that uncertainty estimation can enhance the trustworthiness of AI systems by signaling when predictions might be unreliable. Furthermore, experiments involving combined corruption mechanisms showed that multiple data quality issues can have a far greater impact than isolated sources of corruption. This underscores the importance of robustness testing under realistic operational conditions.

In summary, the study demonstrates that robustness is a crucial prerequisite for sustainable and trustworthy AI systems. Evaluating models solely on error-free datasets is insufficient to understand their real-world performance. Future AI systems should integrate robustness testing, uncertainty estimation, and advanced data quality management techniques to ensure reliable operation in practical environmental monitoring applications.

““latex id=“k7v2as”

REFERENCES

- [1] UCI Machine Learning Repository, “Beijing PM2.5 Air Quality Dataset.” Available: <https://archive.ics.uci.edu/>
- [2] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [3] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.
- [4] G. Ke et al., “LightGBM: A Highly Efficient Gradient Boosting Decision Tree,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [5] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [6] R. J. A. Little and D. B. Rubin, *Statistical Analysis with Missing Data*, 3rd ed., Wiley, 2019.
- [7] S. van Buuren and K. Groothuis-Oudshoorn, “mice: Multivariate Imputation by Chained Equations in R,” *Journal of Statistical Software*, vol. 45, no. 3, pp. 1–67, 2011.
- [8] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning,” *Proc. 33rd International Conference on Machine Learning (ICML)*, 2016.
- [9] European Commission, “Ethics Guidelines for Trustworthy Artificial Intelligence,” High-Level Expert Group on Artificial Intelligence, 2019.

““